R99/A | Fixed target36 vs R89B

Native FP64 winners; same unique full18 VAL checkpoint on TRAIN/VAL/TEST.

Split	Winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	512	153	665	292	0.6148
VAL	4625	26	9	35	16	0.5622
TEST	4598	26	6	25	19	0.5655

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	-0.337654	23255	28178	0.000842089	0.007655
VAL	-0.154321	1318	1443	0.000601406	0.004483
TEST	-0.087654	1364	1435	0.000358003	0.003334

Matched A/B: row-softmax versus fixed64 Sinkhorn, same input-only32 before four instance layers.

Original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35, dropout .1; no new target/loss.

R89B main / R84A secondary historical, neither a newly matched control. Fixed R84 target36.

Not an assignment solver, VRP bound, feasibility test or B-semantics repair.

Zero-count movement alone is not success. Development-exposed TEST; no TEST/LIB selection.

R99/A | Fixed target36 vs R84A

Native FP64 winners; same unique full18 VAL checkpoint on TRAIN/VAL/TEST.

Split	Winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	512	297	82	371	0.6148
VAL	4625	26	16	1	22	0.5622
TEST	4598	26	16	3	22	0.5655

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	0.109602	22455	20857	-0.000241894	-0.003909
VAL	0.008642	1222	1215	0.000284212	0.001665
TEST	0.096296	1248	1170	-0.000014588	-0.001867

Matched A/B: row-softmax versus fixed64 Sinkhorn, same input-only32 before four instance layers.

Original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35, dropout .1; no new target/loss.

R89B main / R84A secondary historical, neither a newly matched control. Fixed R84 target36.

Not an assignment solver, VRP bound, feasibility test or B-semantics repair.

Zero-count movement alone is not success. Development-exposed TEST; no TEST/LIB selection.

R99/A | A Input Transform / Inference Timing

First128 locked TEST input rows/task; synchronized GPU wall time, three repetitions; saved evidence.

Problem	Transform ms/row	Transform + selector ms/row	Cache-hit selector ms/row	Marginal residual
TSP	0.458086	0.858860	0.394064	1.15798
CVRP	0.596066	1.077541	0.478759	1.29677
ATSP	0.012177	0.084773	0.072415	5.3432
OVRP	0.031181	0.101733	0.070387	1.27029
VRPB	0.031275	0.101671	0.070216	1.48684
VRPL	0.031326	0.101197	0.069915	1.36033
VRPTW	0.031272	0.101556	0.070193	1.50045
OVRPTW	0.031680	0.104389	0.073580	1.44011
OVRPB	0.031635	0.102252	0.071098	1.75916
OVRPL	0.031301	0.103080	0.070466	1.26628
VRPBL	0.031294	0.102519	0.070309	1.28209
VRPBTW	0.031127	0.102628	0.070113	1.21418
VRPLTW	0.032597	0.105631	0.070552	1.36161
OVRPBL	0.031346	0.102194	0.070404	1.50991
OVRPBTW	0.031305	0.101744	0.071080	1.28365
OVRPLTW	0.031672	0.101782	0.071823	1.26635
VRPBLTW	0.031608	0.102305	0.070062	1.29172
OVRPBLTW	0.031136	0.102185	0.070369	1.38966

Feature preparation stage wall time: TRAIN/VAL 125.736s; TEST 9.839s (cache reuse allowed).

Original inputs already on GPU; online raw path has no relation_nodes/relation_node_mask.

Transform includes FP16 quantization. All18 tasks include ATSP; no solver decoder.

Excludes dataset I/O and host transfer. Cache-hit selector is not full inference or I/O-inclusive E2E.

Finite64 residuals are recorded, not proof of exact Sinkhorn convergence. Device: NVIDIA GeForce RTX 3090

R99/A | Timing recovery numerical boundary

Private validator revision; approximate messages are NEVER declared bitwise equivalent.

A: Unchanged locked weights/decision rule. Measured path=mixed, TF32 policy=default. Bitwise FP16 equality on first128 rows/task=False; original mixed-online equivalence=False. Full TEST efficacy comes from untouched cached predictions, not a demonstrated exact online run.

TEST efficacy: original locked cached predictions only. No TRAIN/TEST rerun or model selection.

LIB: unchanged original reporter, same checkpoint; its native online computation is disclosed separately.